A Small Generative Model for Image Denoising

A Noise-Gated Conditional GAN with a NAF U-Net Generator

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1. Overview

We submit a **conditional generative adversarial network (cGAN)** that maps a corrupted $256 \times 256 \times 3$ image, together with injected random noise, to a restored $256 \times 256 \times 3$ uint8 image in a **single forward pass**. Different noise draws give different plausible restorations (diversity 0.77 grey levels: mean per-pixel standard deviation of the uint8 outputs over 8 noise seeds), so the model samples from a conditional distribution rather than returning one fixed estimate. On an input of shape $1 \times 3 \times 256 \times 256$ the generator costs **4.98 GFLOPs** with fvcore and **5.09 GMac** with ptflops (both tools count one multiply-add as one operation) and has **5.52 M** parameters; the discriminator is used only during training and adds no inference cost. The model is trained on 26 corruption families, applied singly and in random combinations, over natural, low-light and medical images.

2. Architecture

Generator (NAF U-Net with decoder-only gated noise). The generator is a five-scale U-Net built from NAFBlocks (Chen et al., ECCV 2022). A 3×3 convolution lifts the input to 24 channels. The encoder has four levels at 256, 128, 64 and 32 px with $[2, 2, 2, 4]$ blocks and widths $24, 48, 96, 192$; each level ends with a 2×2 stride-2 convolution that halves the resolution and doubles the channels. A bottleneck of 3 blocks operates at 16 px with 384 channels. The decoder mirrors the encoder with $[1, 1, 1, 2]$ blocks (32 px $\to$ 256 px); each level upsamples with a 1×1 convolution and pixel shuffle and adds the encoder feature map of the same scale (additive skip connection). A final 3×3 convolution predicts a residual that is added to the corrupted input.

- **NAFBlock:** LayerNorm $\to$ 1×1 conv $\to$ 3×3 depthwise conv $\to$ SimpleGate (channel split, element-wise product) $\to$ simplified channel attention $\to$ 1×1 conv, followed by a gated feed-forward branch; both branches are residual with learned zero-initialised scales. The block uses no non-linear activation functions, which keeps it cheap.
- **Noise injection:** after each of the 5 decoder blocks, the feature map x receives $x + s_c \cdot \sigma(\mathrm{conv}_{1\times1}(x)) \cdot n$, where $n \sim \mathcal{N}(0, 1)$ is one noise value per pixel, s_c is a learned per-channel strength (initialised to 0.01) and $\sigma(\mathrm{conv}_{1\times1}(x))$ is a learned per-pixel gate. The encoder and bottleneck are noise-free, so the skip connections carry clean features and the image structure is not perturbed; the gate lets the network learn, per pixel, where variation is added, so that it can be kept to uncertain regions.
- **Inference:** a fresh noise draw is made inside every forward call, so the module measured for FLOPs is exactly the one that runs. The output is clamped to $[0, 1]$, multiplied by 255, rounded and returned as uint8.

Discriminator (training only). A conditional U-Net discriminator with spectral normalisation (as in Real-ESRGAN) gives a real/fake score at every location of its output map. It operates in the wavelet domain: its 18-channel input at 128 px is the three high-frequency Haar bands of the candidate and of the corrupted image, so it judges noise, blur and compression residue (which live in those bands) while colour and brightness are left to the reconstruction losses. Seeing the corrupted image makes it judge whether the output is the right restoration of *that* input.

3. Training objective

For a corrupted image y, clean target x and generator output $\hat{x} = G(y, n)$:

$$\mathcal{L}_G = \underbrace{\mathcal{L}_{\mathrm{Charb}} + 0.05\,\mathcal{L}_{\mathrm{FFT}} + 0.334\,(1 - \mathrm{SSIM}) + 0.1\,\mathcal{L}_{\mathrm{DISTS}} + 0.05\,\mathcal{L}_{\mathrm{wav}}}_{\text{reconstruction, on every output}} + 0.00275\,\mathcal{L}_{\mathrm{adv}} + 4.0\,\max\!\big(0,\ \tau - \tfrac{1}{N}\|\hat{x}_1 - \hat{x}_2\|_1\big).$$

- **Reconstruction terms, one per scored metric:** Charbonnier loss for PSNR; L1 distance between Fourier spectra (FFT) against over-smoothing; $1 - \mathrm{SSIM}$ with the same 7×7 uniform window as `skimage` for SSIM; the DISTS metric itself as a loss (switched on after 50% of training, on 4 images per batch) for perceptual quality; and L1 on Haar sub-bands ($\mathcal{L}_{\mathrm{wav}}$).
- **Adversarial term:** hinge GAN loss, weight 0.00275, ramped in over the first 5k steps. The discriminator is trained with the hinge loss plus a lazy R1 gradient penalty ($\gamma = 1$, every 16 steps).
- **Diversity constraint:** 4 images per batch are restored with a second noise draw; if the two samples differ by less than $\tau = 1.6$ grey levels (mean absolute difference), a hinge penalty pushes them apart, and it stops

acting once the target is met. This keeps the noise input in use, so the model stays generative.

- **Optimisation:** AdamW, learning rate 7.55×10^{-4} (discriminator $0.5\times$), 1k warm-up steps, cosine decay to 10^{-6}, generator weight decay 10^{-4}, gradient clipping 1.0, mixed precision, EMA of the weights (decay 0.999; the EMA weights are submitted). 200,000 steps of 16 images (256×256) plus the 4 extra draws, trained from scratch in about 25 h on one 24 GB GPU.
- **Hyperparameter selection:** learning rate, discriminator learning-rate ratio, adversarial weight, diversity target and SSIM weight were chosen with Optuna (TPE sampler, successive-halving pruning, 38 short trials) under the constraint that diversity stays above 0.6 grey levels.

4. Degradations and training data

Every training pair is generated on the fly: a clean image is corrupted by a random chain of **0 to 4** families (probabilities 12%, 30%, 36%, 17%, 5%; the 12% clean inputs teach the model to leave good images untouched). Each family gets a continuous severity in $[0, 1]$ drawn from $\mathrm{Beta}(1,\ 1.8 + 0.6(k-1))$ for a chain of k families, so longer chains use milder individual corruptions. Chains are applied in physical camera order (optics $\rightarrow$ blur $\rightarrow$ noise $\rightarrow$ compression) or, for 50% of images, in random order. A chain contains at most one family of each type (blur, codec, impulse, ...) except noise, of which up to two may be combined; the brief's nine families carry 65% of the sampling weight. Every corrupted image is quantised to uint8, as at test time.

Group	**Families**
Listed in the brief (9)	Gaussian, Poisson, salt-and-pepper and speckle noise; Gaussian, motion (straight and curved trajectories) and defocus blur; JPEG; chromatic aberration
Additional (13)	spatially correlated, Rician, stripe/banding, sensor (signal-dependent) and periodic noise; random-valued impulses; resize blur; pixelation; ringing (over-sharpening halos); WebP; double JPEG; posterisation; small artifacts (scratches, dead pixels, small blocks)
Domain-specific (4)	demosaicing artifacts and chroma noise (cameras); MRI ghosting; CT ring artifacts

Data. Sources are sampled by weight rather than concatenated, so small domains are not drowned out: *natural* (≈75% of samples: DIV2K, Flickr2K, BSDS500, COCO), *medical* (≈22%: NIH ChestX-ray X-rays, IXI brain MRI slices, MedMNIST CT and ultrasound) and *low-light* (≈3% real LOL images, plus 10% of the natural images darkened synthetically before corruption). Images are resized so that their short side is close to 256 px and randomly cropped to 256×256, with flips, rotations and 4% greyscale conversion. All are public research datasets.

5. Results

Evaluated on our fixed held-out benchmark (396 images: 33 single, combined and unseen corruption cases over natural, low-light and medical images), with `skimage` PSNR/SSIM and `DISTS-pytorch` on uint8 images, as specified in the brief.

Model	PSNR (dB) ↑	SSIM ↑	DISTS ↓	GFLOPs (fvcore) ↓	Diversity*
Submitted model	**32.74**	**0.8740**	**0.1371**	**4.98**	0.77

*Mean per-pixel standard deviation (grey levels, 0–255) of the uint8 outputs over 8 noise seeds, on 96 benchmark images.